Chapter 7 Intro to Networking Exam

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Score: 93% Passing Score: 80%

Question 1: ✔ Correct

Which of the following IP addresses is a valid IP address for a host on a public network?

- 10.3.125.2
- 172.16.254.12
- 142.15.6.1
- 192.168.16.45

EXPLANATION

A public network is a network that does not limit traffic to members of a corporation or other group. The internet is an example of a public network. Certain sets of IP addresses are reserved for private networks only and cannot be used on public networks. They are:

- 10.0.0.0 to 10.255.255.255
- 172.16.0.0 to 172.31.255.255
- 192.168.0.0 to 192.168.255.255

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 2: ✔ Correct

A router is connected to network 192.168.1.0/24 and network 192.168.2.0/24. The router is configured to use RIP and has learned of networks 192.168.3.0/24 and 192.168.4.0/24.

The next hop router for network 192.168.3.0 has changed. You need to make the change with the least amount of effort possible.

What should you do?

- ➡ Wait for convergence to take place.
- Force RIP to perform an immediate update.
- Manually reconfigure the default route to point to the new next hop router.
- Stop and restart the RIP protocol on the router.

EXPLANATION

When using a routing protocol, changes in routing information take some time to be propagated to all routers on the network. The term "convergence" is used to describe the condition when all routers have the same (or correct) routing information.

Static routes in the routing table must be updated manually. Restarting RIP might actually increase the time required for changes to be learned. Forcing an update (if the router supports it) is not a requirement, as the periodic sharing of routes will eventually update the routing table entry.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 3: ✕ Incorrect

This question includes a lab to help you answer the question. **View Lab**

Click the **View Lab** button and use the **tracert** command from the IT Administrator workstation (named **ITAdmin**) to answer the following question.

Below are the IP addresses of the hops in the path between **ITAdmin** and the remote computer on the Internet which has an IP address of **206.8.28.127**.

Drag the IP addresses listed on the left, to the hops described on the right. Use only the addresses that belong to the routers that match the description. Not all of the IP addresses on the right will be used.

Default gateway for ITAdmin	The ISP Internet Router	The PartnetNet Router
~~198.28.56.1~~	~~65.86.24.136~~	✓ 12.34.58.102

172.25.10.1

65.86.1.1

The destination host

✓ 206.8.28.127

Question 4: ✓ Correct

Which of the following routing protocols is classified as a hybrid routing protocol?

- OSPF
- RIP
- IS-IS
- ➡ EIGRP

EXPLANATION

EIGRP is a hybrid routing protocol developed by Cisco for routing within an AS.

RIP is a distance vector protocol, while OSPF and IS-IS are link state protocols.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 5: ✕ Incorrect

You have a router configured to share routing information using RIP. In addition, you have a single static route that identifies a default route for all other networks.

The next hop router for the default route has changed. You need to make the change with the least amount of effort possible. What should you do?

- Force RIP to perform an immediate update.
- ➡ Manually reconfigure the default route to point to the new next hop router.
- Stop and restart the RIP protocol on the router.
- ~~Wait for network convergence to take place.~~

EXPLANATION

With a static route, when changes to the network occur, routing table entries must be modified, added, or removed manually. In this example, the default route was configured manually, so it must be manually updated.

When using a routing protocol, routing changes are made automatically by routers sharing routing information. To make a change for a dynamic route, simply wait for convergence to occur (the term convergence describes the condition when all routers have the same routing information).

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 6: ✔ Correct

You manage a network with multiple subnets connected to the internet.

A user reports that she can't access the internet. You investigate the problem and find that she can access all hosts on the private network, including subnets, but no hosts on the internet.

Which of the following is likely the cause of the problem?

- ➡ Missing default route on a router
- Switching loop on the user's subnet
- Duplex mismatch between a switch and a router
- Incorrect default gateway setting on the computer

EXPLANATION

If you can access all private subnets but not the internet, troubleshoot the connection to the internet. For example, verify that routers have a default route that would apply to internet traffic and make sure the internet link is up and working.

The default gateway on the workstation is configured properly because routing to other subnets works correctly, meaning that the workstation is able to send and receive packets from remote networks.

A duplex mismatch would cause slow communications, but some access might still be possible. A switching loop would affect devices within the subnet, not on other subnets.

REFERENCES

LabSim for Network Pro, Section 7.4.

Question 7: ✔ Correct

Which of the following routing protocols uses paths, rules, and policies instead of a metric for making routing decisions?

- OSPF
- EIGRP
- IS-IS
- RIP
- ➡ BGP

EXPLANATION

BGP is an advanced distance vector protocol (also called a path vector protocol) that uses paths, rules, and policies to make routing decisions instead of a metric.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 8: ✔ Correct

Which of the following best describes OSPF?

- OSPF is a classful distance vector routing protocol.
- OSPF is a classless distance vector routing protocol.
- OSPF is a classful link state routing protocol.
- ➡ OSPF is a classless link state routing protocol.

EXPLANATION

OSPF is a classless link state routing protocol.

RIP version 1 and IGRP are both classful distance vector routing protocols. EIGRP is a hybrid protocol that supports classless addressing.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 9: ✔ Correct

Which of the following is not one of the ranges of IP addresses defined in RFC 1918 that are commonly used behind a NAT server?

- 192.168.0.1 - 192.168.255.254
- ➡ 169.254.0.1 - 169.254.255.254
- 10.0.0.1 - 10.255.255.254
- 172.16.0.1 - 172.31.255.254

EXPLANATION

169.254.0.1 - 169.254.255.254 is the range of IP addresses assigned to Windows DHCP clients if a DHCP server does not assign the client an IP address. This range is known as the Automatic Private IP Addressing (APIPA) range.

The other three ranges listed in this question are defined as the private IP addresses from RFC 1918, which are commonly used behind a NAT server.

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 10: ✔ Correct

You are the network administrator of a branch office of your company. The branch office network is part of a WAN that covers most of the United States. The office has two Windows 2000 servers, two UNIX servers, one Windows NT server, 90 Windows 98 clients, 40 Windows 2000 Professional clients, and five Macintosh clients.

Users have been complaining that they are unable to access resources over the WAN at the main headquarters. You suspect that one of the routers between your office and the main headquarters is not working properly.

What TCP/IP utility can you use to see if a router is working properly?

- **nslookup**
- **nbtstat**
- **netstat**
- ➡ **tracert**

EXPLANATION

tracert shows you the series of routers that are used between the source and destination computers. If a router is not functioning, **tracert** can help you find which router is not working by showing you the last router it was able to contact successfully.

REFERENCES

LabSim for Network Pro, Section 7.4.

Question 11: ✔ Correct

Which of the following tasks do routers perform? (Select two.)

- [x] Maintain information about paths through an internetwork.
- [ ] Multiplex signals onto the same transmission media.
- [ ] Control access to the transmission media.
- [ ] Route data based on hardware device addresses.
- [ ] Identify devices through hardware addresses.
- [x] Route data based on logical network addresses.

EXPLANATION

Routers build and maintain tables of routes through an internetwork and deliver data between networks based on logical network addresses.

REFERENCES

LabSim for Network Pro, Section 7.1.

Question 12: ✔ Correct

What information does the *next hop* entry in a routing table identify?

- ➡ The first router in the path to the destination network.
- The last router in the path to the destination network.
- The number of routers that the packet must go through to reach the destination network.
- A backup router that is used to forward packets addressed to unknown networks.

EXPLANATION

The next hop router is the first (or next) router in the path to the destination network. Each router looks at the destination network in the packet, then consults the routing table to identify the next hop router to the destination network.

The *hop count* identifies the number of routers in the path to the destination network. A default gateway router is a router that is used for packets sent to external networks. Most routers do not have a default gateway setting, but instead use a default route setting, which identifies a next hop router for all unknown networks.

REFERENCES

LabSim for Network Pro, Section 7.1.

Question 13: ✔ Correct

A router is connected to network 192.168.1.0/24 and network 192.168.2.0/24. The router is configured to use RIP and has learned of networks 192.168.3.0/24 and 192.168.4.0/24. The router is also configured with a static route of 0.0.0.0 with a mask of 0.0.0.0.

The router receives a packet addressed to network 10.1.0.0/16. What will the router do with the packet?

- ➡ Forward the packet to the next hop router specified by the route to network 0.0.0.0.
- Send the packet out both of its directly connected networks to the next hop router.
- Drop the packet.
- Send the packet out both of its directly connected networks as a broadcast frame.

EXPLANATION

A route of 0.0.0.0 with a mask of 0.0.0.0 identifies a default route. The default route is used when no other route is a better match. Packets that match no other networks are sent to the next hop router specified by the default route.

REFERENCES

LabSim for Network Pro, Section 7.1.

Question 14: ✔ Correct

Which of the following routing protocols uses relative link cost as the metric?

- RIP
- BGP
- EIGRP
- ➡ OSPF

EXPLANATION

OSPF is a link-state routing protocol used for routing within an AS. OSPF uses relative link cost for the metric.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 15: ✔ Correct

What are the main differences between the OSPF and IS-IS routing protocols?

- OSPF is an IGP routing protocol, while IS-IS is a BGP routing protocol.
- ➡ OSPF requires an area 0, while IS-IS does not.
- OSPF is a classful protocol, while IS-IS is a classless protocol.
- OSPF is a link state protocol, while IS-IS is not.

EXPLANATION

Like OSPF, IS-IS uses areas when designing the network. However, IS-IS does not require an area 0 like OSPF does. Because IS-IS was originally designed for non-IP protocols, it can more easily support IPv6 routing.

Both OSPF and IS-IS have the following characteristics:

- Both are link state protocols.
- Both are classless protocols, supporting CIDR and VLSM.
- Both are Interior Gateway Protocols (IGPs) that are used within an AS.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 16: ✔ Correct

You have a small network at home that is connected to the internet. On your home network, you have a server with the IP address of 192.168.55.199/16. All computers on your home network can connect to the internet.

From your work office, you try to access your home computer using its IP address, but are unable to communicate with the server. You are able to connect to other hosts on the internet.

Why can't you access the server?

- The server must have an entry on a DNS server that exists on the internet.
- ➡ Private addresses are not accessible through the internet.
- The server has been assigned a multicast address.
- The server isn't using the default subnet mask.

EXPLANATION

The server has been assigned a private IP address. Private addresses are not accessible from the internet. Instead, a NAT router translates the private address into a public address, and the public address is used to gain access to the private host.

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 17: ✔ Correct

What is the main difference between RIP and RIPv2?

- ➡ RIP is a classful protocol, while RIPv2 is a classless protocol.
- RIP has a limit of 15 hops, while RIPv2 increases the hop count limit.
- RIP is a distance vector protocol, while RIPv2 is a link state protocol.
- RIP use the hop count for the metric, while RIPv2 uses a relative link cost.

EXPLANATION

RIP v1 is a classful protocol, meaning that the subnet mask is not included in routing updates. With RIP, only the default subnet mask is used to identify networks. RIP v2 is a classless protocol, meaning that the subnet mask is included in routing updates. RIPv2 supports variable length subnet masks (VLSM).

Both RIP and RIPv1 are distance vector protocols and use the hop count for the metric. RIP and RIPv2 have a limit of 15 hops between any two networks.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 18: ✔ Correct

You manage a network with multiple subnets connected to the internet.

A user reports that she can't access the new server used in the accounting department. You check the problem and find out that her computer cannot access any server on that subnet. However, the computer does access other computers on other subnets as well as the internet.

Which of the following is most likely the cause of the problem?

- Incorrect default gateway setting on the computer
- Incorrect VLAN membership of the accounting server
- ➡ Missing route on the default gateway router
- Switching loop on the user's subnet

EXPLANATION

The most likely cause of the problem is a missing route in one of the routers on your internal network (or that the router connecting that subnet to the network is down). Because the problem applies to all hosts in the remote subnet, you can suspect some type of misconfiguration of the router.

The default gateway on the workstation is configured properly because routing to other subnets and the internet works correctly, meaning that the workstation is able to send and receive packets from remote networks. A switching loop would affect devices within the subnet, not on other subnets. Incorrect VLAN membership for the server would affect communications with the server only, not all devices on the subnet (unless all switch ports on the remote subnet are assigned to the same VLAN).

REFERENCES

LabSim for Network Pro, Section 7.4.

Question 19: ✔ Correct

You work for a large multinational organization that has an extensive global network that is interconnected using WAN links and routers.

Lately, users in one location have complained that they are unable to access resources stored on a server named FS23 in a South American branch office.

To troubleshoot the issue, you have done the following:

- Verified that the server is up and running.
- Verified that the various routers in between the two locations are up and running.

You suspect that perhaps one of the routers between the two locations may be dropping packets. To test this theory, you enter the **ping FS23 -f -l 1500** command on your workstation. The ping command returns the following command for each ping packet sent:

"Packet needs to be fragmented but DF set."

What does this mean?

- The destination host is down.
- Collisions are occurring on the network.
- Communications are functioning normally between your workstation and the target server.
- ➡ One of the intermediate routers is an MTU black hole.
- One of the intermediate routers is offline.

EXPLANATION

A black hole router will drop packets when the packet size exceeds the maximum transmission unit (MTU) size. You can use ping to locate a black hole router by setting the following parameters along with the IP address of the remote host:

- **-f** causes the ping utility to send an ICMP echo packet that has the IP do not fragment bit set.
- **-l** sets the buffer (or payload) size of the ICMP echo packet. Specify this size by typing a number after the -l parameter.

The ping test will provide you with helpful information:

- If the MTU of every segment of a routed connection is at least the MTU size, the ping is successful.
- If there is an intermediate segment that has a smaller MTU size, the router returns an ICMP destination unreachable packet, and the ping utility displays a "Packet needs to be fragmented but DF set" message.

REFERENCES

LabSim for Network Pro, Section 7.4.

Question 20: ✔ Correct

Which of the following routing protocols is used by routers on the internet for learning and sharing routes?

- ➡ BGP
- EIGRP
- RIP
- OSPF
- IS-IS

EXPLANATION

BGP is the protocol used on the internet. ISPs use BGP to identify routes between ASs. Very large networks can use BGP internally, but typically only share routes on the internet if the AS has two (or more) connections to the internet through different ISPs.

RIP is used on small private networks, while OSPF and EIGRP are used on larger private networks. IS-IS is used on very large private networks and within the internet service provider (ISP) network.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 21: ✔ Correct

Which of the following techniques allows incoming traffic addressed to a specific port to move through a NAT router and be forwarded to a specific host?

- Port address translation
- EIGRP
- Spanning tree protocol
- ➡ Port forwarding

EXPLANATION

Port forwarding is a type of static NAT implementation where a specific port is mapped to a private IP address. Incoming traffic that is addressed to a specific port is then forwarded to the specified host.

Port address translation (PAT) is used to keep track of which request goes to which host on the internal network. EIGRP is a routing protocol used to exchange information about known routes with other routers. The spanning tree protocol is used to eliminate switching loops within a network.

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 22: ✔ Correct

Which of the following commands would display the output shown here?

```
Route Table
===========================================================================
Interface List
0x1 ........................... MS TCP Loopback interface
0x2 ...00 10 4b 73 0e 0e ...... 3Com 3C90x Ethernet Adapter
===========================================================================
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1   192.168.1.111       1
        127.0.0.0        255.0.0.0        127.0.0.1       127.0.0.1       1
      192.168.1.0    255.255.255.0    192.168.1.111   192.168.1.111       1
    192.168.1.111  255.255.255.255        127.0.0.1       127.0.0.1       1
    192.168.1.255  255.255.255.255    192.168.1.111   192.168.1.111       1
        224.0.0.0        224.0.0.0    192.168.1.111   192.168.1.111       1
  255.255.255.255  255.255.255.255    192.168.1.111   192.168.1.111       1
Default Gateway:       192.168.1.1
===========================================================================
Persistent Routes:
  None
```

- host
- ➡ route print
- ifconfig
- mtr

EXPLANATION

Use the **route print** or **netstat -r** commands to display the contents of the routing table.

The **mtr** command performs a test that is like a combination of the **ping** and **traceroute** commands. The **host** command resolves host names to IP addresses. The **ifconfig** command shows the IP configuration on a Linux computer.

REFERENCES

LabSim for Network Pro, Section 7.4.

Question 23: ✔ Correct

Which of the following routing protocols divides the network into areas, with all networks required to have an area 0 (area 0 identifying the backbone area)?

- RIP
- ➡ OSPF
- IS-IS
- EIGRP

EXPLANATION

OSPF divides a large network into areas. Each autonomous system requires an area 0 that identifies the network backbone. All areas are connected to area 0, either directly or indirectly through another area. Routes between areas must pass through area 0.

IS-IS uses areas, but does not have an area 0 requirement. Neither RIP nor EIGRP use areas.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 24: ✔ Correct

Examine the following output:

```
4 22 ms 21 ms 22 ms sttlwa01gr02.bb.ispxy.com [154.11.10.62]
5 39 ms 39 ms 65 ms plalca01gr00.bb.ispxy.com [154.11.12.11]
6 39 ms 39 ms 39 ms Rwest.plalca01gr00.bb.ispxy.com [154.11.3.14]
7 40 ms 39 ms 46 ms svl-core-03.inet.ispxy.net [205.171.205.29]
8 75 ms 117 ms 63 ms dia-core-01.inet.ispxy.net [205.171.142.1]
```

Which of these commands produced this output?

- **nslookup**
- ➡ **tracert**
- **pingroute**
- **ping**

EXPLANATION

The output is from a **tracert** command run on a Windows Server 2003 system. The **tracert** command provides information on each step in the route a packet takes to reach a remote host. Responses from each hop on the route are measured three times to provide an accurate representation of how long the packet takes to reach and be returned by that host. This information can be useful for locating congestion points on a network or when verifying that network routing is operating as expected.

The **ping** command is used to test connectivity between devices on a network. Like **tracert**, **ping** sends three packets to the target host, but it does not report information on any intermediate devices it traverses to reach the target.

nslookup is a tool provided on Linux, Unix, and Windows systems that allows manual name resolution requests to be made to a DNS server. This can be useful when troubleshooting name resolution problems.

REFERENCES

LabSim for Network Pro, Section 7.4.

Question 25: ✔ Correct

You have a computer that is connected to the internet through a NAT router. You want to use a private addressing scheme for your computer. Which of the following IP addresses could you assign to the computer? (Select all that apply.)

- [ ] 240.12.188.1
- [ ] 127.0.0.1
- [x] 192.168.12.253
- [ ] 224.15.166.12
- [ ] 32.188.99.10
- [x] 172.18.188.67
- [x] 10.0.12.15

EXPLANATION

Of the addresses listed here, the following are in the private IP address ranges:

- 10.0.12.15 (private range = 10.0.0.0 to 10.255.255.255)
- 172.18.188.67 (private range = 172.16.0.0 to 172.31.255.255)
- 192.168.12.253 (private range = 192.168.0.0 to 192.168.255.255)

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 26: ✔ Correct

Which of the following are true of the IS-IS routing protocol? (Select two.)

- [ ] A router is the boundary between one area and another.
- [ ] It uses bandwidth and delay for the metric.
- [ ] It is a classful protocol.
- [ ] It is best suited for small networks.
- [x] It supports IPv6 routing.
- [x] It divides large networks into areas.

EXPLANATION

IS-IS is a link-state routing protocol used for routing within an AS. IS-IS is classless and uses relative link cost for the metric. Large networks are divided into areas. A network link is the boundary between one area and another area. IS-IS is best suited for large private networks.

REFERENCES

LabSim for Network Pro, Section 7.2.

Question 27: ✔ Correct

You are the network administrator for a small company that implements NAT to access the internet. You recently acquired five servers that must be accessible from outside your network. Your ISP has provided you with five additional registered IP addresses to support these new servers, but you don't want the public to access these servers directly. You want to place these servers behind your firewall on the inside network, yet still allow them to be accessible to the public from the outside.

Which method of NAT translation should you implement for these five servers?

- Overloading
- ➡ Static
- Dynamic
- Restricted

EXPLANATION

Static translation consistently maps an unregistered IP address to the same registered IP address on a one-to-one basis. Static NAT is particularly useful when a device needs to be assigned the same address so it can be accessed from outside the network, such as web servers and other similar devices.

Dynamic translation would not work for these servers because it maps an unregistered host IP address to any available IP address configured in a pool of one or more registered IP addresses. Accessing a server assigned one of these addresses would be nearly impossible because the addresses are still shared by multiple hosts.

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 28: ✔ Correct

Which of the following associates a port number with a host on a private network?

- CIDR
- ➡ PAT
- NAT
- VLSM

EXPLANATION

Port address translation (PAT) associates a port number with the translated address. Use PAT to allow multiple private hosts to share a single public address. Each private host is associated with a unique port number.

Technically speaking, NAT translates one address to another. With only NAT, you would have to have a public address for each private host. NAT would associate a single public address with a single private address. Because virtually all NAT routers perform port address translation, most routers that are configured with NAT are really performing PAT. When you use a NAT router, you normally use PAT, not just NAT. (NAT is typically used synonymously with PAT.)

Classless inter-domain routing (CIDR) allows non-default subnet masks (variable-length subnet masks, or VLSMs).

REFERENCES

LabSim for Network Pro, Section 7.3.

Question 29: ✔ Correct

A router is connected to network 192.168.1.0/24 and network 192.168.2.0/24. The router is configured to use RIP and has learned of networks 192.168.3.0/24 and 192.168.4.0/24. There is no default route configured on the router.

The router receives a packet addressed to network 10.1.0.0/16. What will the router do with the packet?

- Hold the packet in cache until a matching route is learned or configured.
- ➡ Drop the packet.
- Send the packet out both of its directly-connected networks as a broadcast frame.
- Send the packet to both networks 192.168.3.0 and 192.168.4.0 and to the next hop router.

EXPLANATION

If a packet does not match any route in a routing table, the router drops the packet. In this example, the router does not know about the destination network, and it is not configured with a default route. With a default route, the router will forward the packet to the next hop router specified by the default route.

REFERENCES

LabSim for Network Pro, Section 7.1.

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